

BOONE, IA, USA

WMO: 725486

Lat: 42.049N

Lon: 93.848W

Elev: 1160

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-0.1	-13.1	2.8	-5.5	-8.4	3.7	0.3	29.7	15.6	26.6	19.6	9.6	340	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	91.0	76.3	88.9	75.3	86.1	73.5	80.1	87.4	78.0	84.9	75.9	83.2	10.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	149.3	84.9	75.4	139.0	82.4	73.2	128.9	80.5	44.6	87.2	42.4	85.5	40.2	83.4	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.3	23.5	20.4	-13.5	95.2	7.2	2.9	-18.7	97.3	-22.9	98.9	-26.9	100.6	-32.1	102.7		
			-13.1	83.0	6.7	2.0	-17.9	84.4	-21.8	85.6	-25.5	86.7	-30.4	88.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	21.8	25.8	36.9	49.7	61.1	70.7	74.1	72.3	64.9	52.2	38.6	26.8
	DBStd	20.84	12.49	12.41	12.56	9.67	8.80	6.50	5.56	5.24	7.96	9.70	11.07	11.18
	HDD50	3302	875	680	436	120	12	0	0	0	3	89	368	719
	HDD65	6475	1340	1099	871	466	182	23	3	6	98	412	792	1183
	CDD50	3192	0	2	31	111	357	620	747	690	450	157	26	1
	CDD65	889	0	0	1	7	62	192	286	231	95	15	0	0
	CDH74	7892	0	0	12	119	599	1710	2596	1833	856	158	9	0
	CDH80	2613	0	0	0	26	190	587	925	579	262	43	1	0
(14) Wind	WSAvg	9.8	10.8	11.0	11.7	12.2	11.1	8.8	6.9	6.5	7.8	9.8	10.5	10.4
(15) Precipitation	PrecAvg	35.9	1.0	1.1	2.3	3.7	4.9	5.1	5.0	4.8	3.0	2.1	1.9	1.1
	PrecMax	55.1	2.0	1.9	5.2	8.0	8.5	11.4	14.6	11.2	6.9	5.7	4.4	2.2
	PrecMin	24.8	0.4	0.1	0.1	0.8	1.3	0.8	1.6	0.9	1.0	0.3	0.0	0.0
	PrecStd	6.9	0.4	0.6	1.4	2.0	2.0	2.7	3.1	2.6	1.5	1.3	1.3	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	61.3	75.4	83.6	90.5	92.7	94.9	93.4	90.9	85.9	74.2	59.1
		MCWB	43.4	52.3	60.4	64.5	70.4	75.8	80.2	76.6	72.6	70.0	58.6	53.5
	2%	DB	46.1	53.7	68.0	77.1	84.5	90.0	91.0	89.8	87.3	78.5	65.9	51.5
		MCWB	39.9	47.8	56.8	60.3	67.3	75.1	78.2	77.2	71.7	64.0	55.2	46.4
	5%	DB	41.3	46.3	61.4	71.9	80.9	86.4	88.4	85.9	82.2	72.8	60.5	45.7
		MCWB	37.1	42.0	54.4	57.1	65.2	72.4	77.1	75.7	69.0	61.2	51.6	41.1
	10%	DB	36.8	41.3	55.0	66.2	76.3	83.1	85.7	82.3	79.3	68.1	54.9	40.9
		MCWB	34.2	37.6	48.4	54.4	63.2	71.0	75.5	73.2	67.0	58.5	47.7	37.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	53.8	63.1	66.9	74.3	80.5	83.8	82.4	77.0	71.5	60.3	55.0
		MCDB	52.8	59.7	72.1	78.3	84.2	88.2	91.7	90.0	84.4	79.7	69.4	58.4
	2%	WB	40.9	47.5	58.9	62.6	70.6	77.8	80.8	79.2	74.7	67.2	57.0	47.3
		MCDB	46.1	51.9	64.7	72.8	81.3	85.8	88.2	86.1	82.7	75.0	64.3	49.1
	5%	WB	37.5	42.5	54.6	59.3	67.9	75.1	78.6	77.0	72.6	63.6	52.6	41.4
		MCDB	41.2	47.0	61.3	68.6	76.3	83.1	85.4	83.8	79.7	69.4	59.2	45.7
	10%	WB	34.4	38.2	49.1	56.0	65.6	73.0	76.6	74.7	70.0	59.5	48.1	37.2
		MCDB	36.9	41.3	55.3	64.5	73.4	80.6	83.3	81.1	76.0	66.5	54.5	40.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	15.5	17.6	20.5	19.5	18.8	18.4	18.0	21.4	20.0	17.3	14.4
		MCDBR	20.1	21.1	26.3	30.8	25.3	22.1	20.0	20.9	25.1	26.7	25.8	19.7
		MCWBR	15.2	15.6	16.6	16.6	11.9	11.0	11.0	11.8	12.5	14.5	16.0	15.4
	5% WB	MCDBR	18.9	20.8	24.6	24.9	21.3	19.4	18.3	18.6	20.6	21.5	23.4	18.4
		MCWBR	14.6	15.9	16.4	15.8	12.1	11.8	12.0	12.1	11.9	14.6	16.2	15.2
(40) Clear-Sky Solar Irradiance	taub		0.285	0.302	0.323	0.379	0.408	0.421	0.435	0.419	0.388	0.341	0.297	0.284
	taud		2.421	2.378	2.421	2.284	2.273	2.282	2.223	2.275	2.318	2.443	2.507	2.446
	Ebn at Noon		275	287	294	282	276	271	266	267	267	269	269	266
	Edh at Noon		24	30	33	40	42	42	44	40	36	28	23	22
(44) All-Sky Solar Radiation	RadAvg		604	925	1202	1506	1780	1977	2013	1731	1434	963	653	495
	RadStd		60	67	117	125	135	174	133	116	102	118	51	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page